

SOPOCHNAYA KARGA, Russian Federation

WMO: 208710

Lat: 71.875N

Lon: 82.706E

Elev: 5

StdP: 101.26

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.5	-40.6	-45.1	0.0	-42.1	-43.2	0.1	-40.5	17.3	-20.6	16.0	-17.9	3.1	90	1.102

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.7	19.7	15.8	17.6	14.6	15.7	13.2	16.3	19.1	15.0	17.4	13.3	15.5	4.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	10.7	17.7	13.5	9.6	16.8	12.0	8.7	15.0	45.8	19.1	42.0	17.3	37.5	15.6	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.5	14.5	13.1	-43.2	22.4	3.3	2.1	-45.6	23.9	-47.5	25.2	-49.4	26.4	-51.8	27.9		
			-43.2	18.0	2.6	1.9	-45.1	19.4	-46.6	20.6	-48.1	21.6	-50.0	23.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.9	-28.5	-26.1	-21.0	-15.6	-7.0	4.0	11.0	8.5	4.0	-6.6	-18.7	-23.7
	DBStd	15.10	7.99	8.47	9.27	7.43	5.13	4.05	4.42	3.35	3.67	5.93	7.32	8.33
	HDD10.0	7360	1194	1012	960	768	526	186	42	69	184	514	861	1044
	HDD18.3	10298	1452	1245	1218	1018	784	429	229	304	431	773	1111	1302
	CDD10.0	106	0	0	0	0	0	7	73	23	3	0	0	0
	CDD18.3	3	0	0	0	0	0	0	3	0	0	0	0	0
	CDH23.3	4	0	0	0	0	0	0	4	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.3	6.1	6.0	6.3	5.9	6.4	5.4	6.0	6.5	6.6	6.9	6.2	6.9
Precipitation	PrecAvg	670	62	56	51	47	50	39	43	58	56	78	66	64
	PrecMax	850	152	99	85	114	107	73	107	98	114	119	113	126
	PrecMin	343	7	11	11	0	14	11	3	15	16	19	10	13
	PrecStd	109	36	26	20	23	21	15	27	20	22	23	28	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.6	-3.3	-0.4	1.2	2.2	18.9	23.8	19.0	13.9	5.0	-1.7	-2.6
		MCWB	-2.9	-3.5	-1.0	0.3	1.2	14.7	17.6	15.3	11.8	4.1	-2.1	-3.0
	2%	DB	-9.2	-7.1	-2.4	-0.4	1.4	14.6	21.3	16.8	12.1	3.1	-3.4	-4.5
		MCWB	-9.5	-7.4	-2.7	-0.8	0.6	11.9	16.7	14.7	10.7	2.3	-3.7	-4.7
	5%	DB	-14.5	-10.3	-4.6	-2.2	0.7	12.1	19.4	14.9	10.6	1.9	-5.7	-7.7
		MCWB	-14.8	-10.6	-4.9	-2.5	0.2	9.9	15.7	13.0	9.3	1.3	-6.1	-8.1
	10%	DB	-17.0	-13.4	-7.0	-4.4	0.1	10.2	17.6	13.6	9.2	0.8	-8.1	-11.1
		MCWB	-17.3	-13.7	-7.4	-4.8	-0.4	8.4	14.6	11.8	8.0	0.2	-8.6	-11.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.8	-3.4	-0.7	0.5	1.6	14.8	18.9	16.0	12.1	4.5	-2.0	-3.0
		MCDB	-2.4	-3.2	-0.3	1.2	2.2	17.9	22.4	17.6	13.4	5.0	-1.7	-2.8
	2%	WB	-9.5	-7.4	-2.7	-0.8	0.8	11.9	17.2	14.8	10.7	2.4	-3.8	-4.8
		MCDB	-9.1	-7.1	-2.5	-0.4	1.2	13.8	20.0	16.5	12.0	2.8	-3.5	-4.6
	5%	WB	-14.8	-10.6	-4.9	-2.5	0.3	10.0	16.1	13.3	9.5	1.4	-6.1	-8.0
		MCDB	-14.6	-10.4	-4.6	-2.2	0.7	11.9	19.2	14.7	10.5	1.9	-5.8	-7.7
	10%	WB	-17.2	-13.7	-7.4	-4.8	-0.4	8.4	14.9	12.0	8.1	0.3	-8.5	-11.3
		MCDB	-16.9	-13.4	-7.0	-4.4	0.1	10.2	17.5	13.2	9.1	0.9	-8.1	-11.0
Mean Daily Temperature Range	5% DB	MDBR	6.2	7.1	7.6	8.5	5.1	4.0	5.7	4.3	3.4	4.4	5.8	6.5
		MCDBR	9.5	9.2	8.1	9.2	3.9	6.9	6.9	5.5	4.0	2.7	8.1	8.9
		MCWBR	9.4	9.1	8.2	9.2	3.8	5.2	4.6	4.1	3.4	2.6	8.1	8.8
	5% WB	MCDBR	9.6	8.8	8.2	8.9	3.9	6.4	6.5	4.9	3.9	3.0	8.2	8.8
		MCWBR	9.5	8.7	8.3	8.9	3.8	5.2	4.7	3.9	3.4	3.0	8.2	8.7
Clear-Sky Solar Irradiance	taub	N/A	0.219	0.291	0.314	0.338	0.349	0.373	0.358	0.308	0.215	N/A	N/A	
	taud	N/A	1.925	2.012	2.067	2.093	2.307	2.345	2.366	2.398	2.030	N/A	N/A	
	Ebn at Noon	0	388	647	786	819	830	789	749	674	391	0	0	
	Edh at Noon	0	34	80	109	123	104	97	83	57	28	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.17	1.49	3.87	5.75	5.61	4.84	2.96	1.30	0.33	0.00	0.00	
	RadStd	0.00	0.03	0.16	0.34	0.32	0.39	0.61	0.46	0.14	0.05	0.00	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (1 neighbor)	+1.59	+1.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page